Computer Core & Data Network: System Design Description

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Revision	C
Status	APPROVED
Classification	INTERNAL
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1. System overview

The computer core and data network connect every station system. Maintenance terminals and data ports give technicians and service droids access to the network.

The system is led by Computing & Communications. It is delivered through 6 work packages with a combined budget of 76,375.12 Mcr.

2. Components

Code	Component	Specification	Drawing	Bill of materials
CN-CCO	Central computer core	SHV-CN-SPEC-0101	SHV-CN-DWG-0201	SHV-CN-BOM-0301
CN-DNB	Data network backbone	SHV-CN-SPEC-0102	SHV-CN-DWG-0202	SHV-CN-BOM-0302
CN-MDP	Maintenance data port	SHV-CN-SPEC-0103	SHV-CN-DWG-0203	SHV-CN-BOM-0303
CN-MTT	Maintenance terminal	SHV-CN-SPEC-0104	SHV-CN-DWG-0204	SHV-CN-BOM-0304

3. Architecture

3.1 Central computer core

The central computer core is specified in SHV-CN-SPEC-0101. Its governing parameter is the processing nodes, set at 5,485 ea. The responsible engineer is Riven Fairholm.

3.2 Data network backbone

The data network backbone is specified in SHV-CN-SPEC-0102. Its governing parameter is the backbone links, set at 234 ea. The responsible engineer is Vira Renner.

3.3 Maintenance data port

The maintenance data port is specified in SHV-CN-SPEC-0103. Its governing parameter is the port count, set at 12,761 ea. The responsible engineer is Dorne Lancer.

3.4 Maintenance terminal

The maintenance terminal is specified in SHV-CN-SPEC-0104. Its governing parameter is the terminal count, set at 4,077 ea. The responsible engineer is Neve Daventry.

4. Interfaces

Document	Title
SHV-ICD-HR-CN	Interface Control Document: Hypermatter Reactor / Computer Core & Data Network

Document	Title
SHV-ICD-HB-CN	Interface Control Document: Docking & Hangar Bays / Computer Core & Data Network
SHV-ICD-CN-CM	Interface Control Document: Computer Core & Data Network / Communications Relay
SHV-ICD-CN-OC	Interface Control Document: Computer Core & Data Network / Operations Centre
SHV-ICD-CN-TR	Interface Control Document: Computer Core & Data Network / Internal Transit
SHV-ICD-RL-CN	Interface Control Document: Research Laboratories / Computer Core & Data Network

5. Work packages

Work package	Name	Owner	Status	Complete
WP-SHV-CN-01	Computer Core & Data Network: Design & analysis	Aldo Palin	IN_PROGRESS	75.0%
WP-SHV-CN-02	Computer Core & Data Network: Fabricate & install central computer core	Rafe Craddock	IN_PROGRESS	73.3%
WP-SHV-CN-03	Computer Core & Data Network: Fabricate & install data network backbone	Ursel Marrek	IN_PROGRESS	99.5%
WP-SHV-CN-04	Computer Core & Data Network: Fabricate & install maintenance data port	Ursel Marrek	IN_PROGRESS	75.6%
WP-SHV-CN-05	Computer Core & Data Network: Fabricate & install maintenance terminal	Aldo Palin	IN_PROGRESS	93.0%
WP-SHV-CN-06	Computer Core & Data Network: Integration & test	Carys Easton	IN_PROGRESS	38.1%

6. Risks

- Supply of central computer core assemblies depends on a single supplier.
- Supply of data network backbone assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.